

Choosing Between Two Virtual Amplifiers —Guitarix and Guitar Rig

Given the high cost of audio equipment required to amplify and process the sound produced by musical instruments like guitars, consider opting for a virtual amplifier and sound processor like Guitarix, which is free and open source, or its paid version, Guitar Rig.



MIDI (Musical Instrument Digital Interface) is a technical standard used for communication between audio devices. It helps in transferring data about the pitch, notations, velocity, etc., of music being transferred from one instrument to another. It can be used for the manipulation of live sound with no notable latency. MIDI makes it possible for anyone to create music even without using instruments.

In 1999, members of the USB-IF developed a standard for MIDI over USB. This is the most used form of MIDI in the world of electronic live music today because of its very low power consumption and portability.

The JACK audio connection kit (or JACK) is an open source professional sound server daemon that helps in reducing the latency (the delay between the production and playback of the sound signals). It usually uses hardware back-ends like ALSA, PortAudio, OSS, etc. JACK was originally implemented in C and a later update, JACK 2, was implemented in C++. JACK is available on almost any OS ranging from FreeBSD and Linux to Mac and Windows.

The need for virtual amplification

Anyone who plays a guitar knows that the equipment required to amplify the sound or synthesise it is expensive.

After buying a costly guitar, there's little cash left for a huge head amplifier unit, its extension cabinet and foot switches. Then there are the issues of ground noise and power supply to be addressed. Virtual amplification provides a cheap solution for the above problems by substituting the expensive amplifiers or guitar pedals and modelling the sound of the guitar. Various Vats (virtual instruments) let the computer mimic the sounds of expensive guitar pedals and amps. The equipment used is also less bulky as you need to carry around just a laptop, an audio interface and a bunch of cables along with your precious guitar.

Introducing Guitarix and Guitar Rig

Guitarix is a free open source virtual guitar amplifier, which runs using the JACK audio connection kit. It is available for most UNIX based systems. It provides 25 built-in modules for you to modify your guitar's sound and that too in stereo. The latency on this setup can go as low as 10ms.

Guitar Rig is paid-for virtual guitar amplifier and modelling software developed by the infamous Native Instruments. It is well known as a standalone amplifier for performances and also as a virtual guitar effects processor for studio recording applications.